

8.4 Station 12 feeder restoration and outage isolation

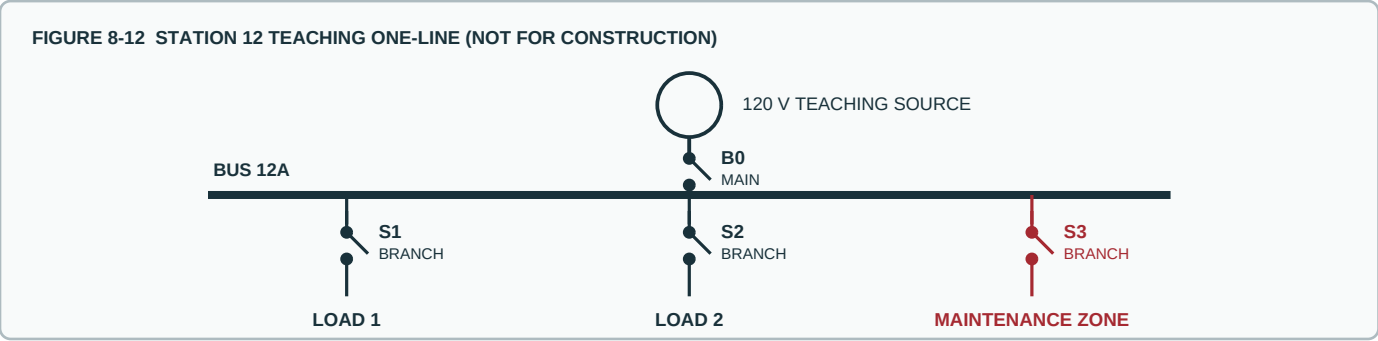
Functional design specification excerpt

Document ID	RBMP-FDS-DA-004	System	Station 12 distribution automation
Revision	4	Effective date	2026-08-14
Owner	OT Systems Engineering	Approval	Operations / Protection Engineering
Excerpt scope	Sections 8.4-8.7	Classification	Internal training copy

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8.4.1 Purpose

This section defines the close-permission behavior for main breaker B0 and branch switching devices S1 through S3 during planned maintenance and restoration. It specifies the operating properties to be preserved by the controller. Protection settings, field switching procedures, communications security, and personnel authorization remain governed by their owning documents.



8.4.2 Operating context

The normal training scenario declares S3 out of service while S1 and S2 remain required for service. B0 may be open while branch contacts retain their last position. Consequently, a zero source-current measurement does not establish that a later B0 close will be safe. The controller shall evaluate the state that would result from the requested operation.

8.5 Required controller behavior

The words SHALL and SHALL NOT identify mandatory behavior in this excerpt.

ID	Requirement	Rationale / owner
FDS-DA-401	A close operation SHALL NOT result in any device declared out of service becoming energized.	Primary safety property. Operations owns the approved outage declaration.
FDS-DA-402	The close-permission function SHALL evaluate the resulting contact topology: ResultingPositions = ActualPositions OR TargetMask.	Covers upstream closes that energize an already-closed downstream branch.
FDS-DA-403	A close operation SHALL be denied while the independent trip latch is set.	Protection state constrains restoration; it is not cleared by this function.
FDS-DA-404	A close decision SHALL fail closed when required position or outage-state evidence is stale, invalid, or from a different command generation.	Freshness/versioning signal is an interface requirement. This excerpt does not define its transport.
FDS-DA-405	Present current MAY support diagnosis but SHALL NOT be used alone to infer the safety of a topology-changing close.	Current before a close cannot prove which downstream section the close will energize.
FDS-DA-410	With the out-of-service branch isolated, the design SHALL permit a bounded restoration sequence that supplies all required healthy branches.	Service property prevents a deny-all implementation from satisfying the design.
FDS-DA-411	A protocol acknowledgement or controller permit SHALL NOT be reported as actual device movement. Position feedback shall originate from the device/process boundary.	Separates request, decision, actuation, and physical consequence.

8.5.1 Close-permission sequence

Step	Owning function	Required behavior
1	Request adapter	Validate command form and assign a nonzero request sequence. Do not actuate a device.
2	State adapter	Present current positions, target mask, approved out-of-service mask, trip state, and measurement context.
3	Compiled controller	Evaluate FDS-DA-401 through FDS-DA-405 for the same request generation; return permit or deny.
4	Device adapter	Apply only a current permit. Timeout, controller stop, or sequence mismatch shall fail closed.
5	Process / feedback	Derive actual contact position and electrical measurements independently of the requested value.

Boundary note: This FDS establishes functional behavior. It does not establish requester identity, network authenticity, field isolation, relay coordination, or compliance with any named protocol or standard.

8.6 Point contract and source authority

The following integration points are shown for the training implementation. Register locations are configuration details, not proof of physical state or authority.

Point	Location	Owner	Meaning / limitation
RequestSeq	%MW0 / 1024	Request adapter	Nonzero command generation; binds one request to one decision.
ActualPositions	%MW1 / 1025	Device adapter	Bit mask B0=01, S1=02, S2=04, S3=08. Requires independent freshness evidence.
TargetMask	%MW2 / 1026	Request adapter	One requested close target; not an actual device position.
OutOfServiceMask	%MW3 / 1027	Outage record adapter	Approved unavailable section mask. A local label alone is not authority.
TripLatched	%MW4 / 1028	Protection adapter	Independent protection state; 0 clear, 1 latched.
PresentCurrent_dA	%MW5 / 1029	Measurement adapter	Source current in deciamps. Observation only; not a topology guarantee.
CurrentLimit_dA	%MW6 / 1030	Configuration	Teaching threshold. Does not replace FDS-DA-401/402.
DecisionCode	%MW7 / 1031	Compiled controller	0 deny, 1 permit for DecisionSeq.
DecisionSeq	%MW8 / 1032	Compiled controller	Request generation associated with current decision.
Heartbeat	%MW9 / 1033	Compiled controller	Scan liveness only; not proof of correct logic or safe actuation.

8.6.1 Source-use rules

Source	Authoritative for	Cannot establish by itself
This FDS excerpt	Required functional behavior and ownership boundaries	Actual field state, current implementation, or test result
Approved outage record	Which section is declared unavailable for the active generation	Contact position, de-energization, or isolation completeness
Controller source/build	Implemented decision logic and artifact identity	That the device moved or the process remained safe
Device/process telemetry	Reported contact and electrical consequence at a stated time	Requester authority or correctness of controller logic
Captured regression	Observed behavior for the named build, scenario, and run	Untested states or operation outside the stated assumptions

Traceability rule: Claims shall name the requirement, controller build, run identifier, evidence source, and assumptions. Unknown or unavailable evidence shall remain explicit; it shall not be inferred from a communication success or a plausible label.

8.7 Verification and acceptance excerpt

Acceptance requires both a prohibited-transition test and a legitimate-service test against the same compiled controller build. Independent process evidence shall determine the physical result.

Test	Initial state and request	Required result	Evidence retained
AT-DA-17 Upstream close	B0 open; S3 contact closed; S3 declared OOS; source current 0 A. Request CLOSE B0.	DENY. B0 remains open; S3 remains de-energized; no protection trip.	Request/decision generation, build ID, actual contacts, bus voltage, branch current, trip state.
AT-DA-18 Healthy restoration	B0 open; S1/S2/S3 open; S3 declared OOS. Request CLOSE S1, CLOSE S2, then CLOSE B0.	PERMIT valid sequence. S1 and S2 are supplied; S3 remains isolated; no trip.	Three requests/decisions, actual positions, sustained-service result, measurements.
AT-DA-19 Relocated outage	B0 open; S2 declared OOS; S1 and S3 required. Perform bounded restoration.	S1 and S3 may be supplied while S2 remains de-energized. No target-specific exception is acceptable.	Scenario ID, outage mask, build ID, resulting topology, service and safety verdicts.
DR-DA-20 Stale position review	A previously safe contact report expires or physical state changes before actuation.	Do not extend a prior permit. Revalidate at the final actuation boundary or fail closed.	Freshness/version mechanism is not supplied by the current fixture; record this as a bounded claim limitation.

8.7.1 Acceptance conditions

- A controller that blocks every close fails FDS-DA-410 even if no unsafe energization occurs.
- A controller that blocks only direct close commands to the OOS branch fails AT-DA-17.
- A protective trip after energization does not satisfy FDS-DA-401; protection is an independent layer.
- A request acknowledgement, decision code, or HMI echo does not satisfy actual-position evidence.
- Results shall be labeled TESTED, REPLAYED, or PROPOSED according to how the evidence was produced.

8.7.2 Revision note

Rev	Date	Description	Approved by
3	2026-07-29	Separated protocol acceptance from device-position confirmation.	OT Systems / Operations
4	2026-08-14	Added resulting-topology requirement and stale-evidence design review.	OT Systems / Protection

END OF SUPPLIED EXCERPT. Referenced but not supplied: RBMP-OPS-SW-012 switching procedure; RBMP-PRT-012 protection settings; RBMP-SEC-006 remote-access standard; Station 12 as-built drawings. Students shall not invent facts from these absent sources.

Course provenance

Prepared for CS 6494 as a fictional internal-documentation excerpt. Functional identifiers align with the Substation Recovery Lab, but the document makes no claim of standards conformance, vendor equivalence, or use at a real facility.

Document identity: RBMP-FDS-DA-004, Rev 4, excerpt pages 173-176.